

Variants of Loops in C

1.Variants of the **for** Loop

Standard **for** Loop

```
for(int i = 0; i < 10; i++) {  
    printf("%d\n", i);  
}
```

Reverse Counting Loop

```
for(int i = 10; i >= 1; i--) {  
    printf("%d\n", i);  
}
```

Multiple Variables

```
for(int i = 0, j = 10; i < j; i++, j--) {  
    printf("i = %d, j = %d\n", i, j);  
}
```

Infinite **for** Loop

```
for(;;) {  
    printf("Running...\n");  
}
```

Omitting Initialization

```
int i = 0;  
for(; i < 5; i++) {  
    printf("%d\n", i);}
```

Omitting Condition

```
for(int i = 0; ; i++) {  
    if(i == 5) break;
```

```
    printf("%d\n", i);  
}
```

Omitting Increment

```
for(int i = 0; i < 5; ) {  
    printf("%d\n", i);  
    i++;  
}
```

Null **for** Loop

```
for(int i = 0; i < 1000000; i++);
```

Using **continue**

```
for(int i = 1; i <= 5; i++) {  
    if(i == 3) continue;  
    printf("%d\n", i);  
}
```

Using **break**

```
for(int i = 1; i <= 5; i++) {  
    if(i == 4) break;  
    printf("%d\n", i);  
}
```

Floating Counter

```
for(float x = 0.0; x <= 1.0; x += 0.2) {  
    printf("%.1f\n", x);  
}
```

2. Variants of the **while** Loop

Standard **while** Loop

```
int i = 0;
while(i < 5) {
    printf("%d\n", i);
    i++;
}
```

Infinite **while** Loop

```
while(1) {
    printf("Running...\n");
}
```

Using **break**

```
int i = 0;
while(i < 10) {
    if(i == 5) break;
    printf("%d\n", i);
    i++;
}
```

Using **continue**

```
int i = 0;
while(i < 5) {
    i++;
    if(i == 3) continue;
    printf("%d\n", i);
}
```

Without Initialization Inside Loop

```
int i = 10;
while(i >= 1) {
    printf("%d\n", i);
    i--;
}
```

Input Validation

```
int x;
scanf("%d", &x);
while(x <= 0) {
    printf("Enter again: ");
    scanf("%d", &x);
}
```

Complex Conditions

```
int a = 5, b = 20;
while(a < 10 && b > 15) {
    printf("%d %d\n", a, b);
    a++; b--;
}
```

Loop That Never Runs

```
int n = 0;
while(n > 0) {
    printf("Won't run\n");
}
```

3. Variants of the **do-while** Loop

Standard **do-while** Loop

```
int i = 0;
do {
    printf("%d\n", i);
    i++;
} while(i < 5);
```

Loop That Runs At Least Once

```
int x = 10;
do {
    printf("This prints once even if condition is false.\n");
} while(x < 5);
```

Using **break**

```
int i = 0;
do {
    if(i == 3) break;
    printf("%d\n", i);
    i++;
} while(i < 10);
```

Using **continue**

```
int i = 0;
do {
    i++;
    if(i == 3) continue;
```

```
    printf("%d\n", i);  
} while(i < 5);
```

Input Validation Version

```
int n;  
do {  
    printf("Enter a positive number: ");  
    scanf("%d", &n);  
} while(n <= 0);
```

Infinite **do-while Loop**

```
do {  
    printf("Running...\n");  
} while(1);
```